



Exact and numerical solutions for non-linear Burger's equation by VIM

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ABSTRACT

In this article variational iteration method (VIM), established by He in (1999), is considered to solve nonlinear Burger's equation. This method is a powerful tool for solving a large number of problems. Using variational iteration method, it is possible to find the exact solution or a closed approximate solution of a problem. Comparing the results with those of Adomian's decomposition and finite difference methods reveals significant points. To illustrate the ability and reliability of the method, some examples are provided.

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1. Introduction

Burger's equation has attracted much attention. Solving this equation has been an interesting task for mathematicians. This equation has been found to describe various kinds of phenomena such as a mathematical model of turbulence and approximate theory of flow through a shock wave traveling in a viscous fluid [1,2]. Numerical methods which are commonly used such as finite difference or characteristics method need a large amount of computation and usually the effect of round-off error causes the loss of accuracy. Analytical methods for solving Burger's equation are very restricted and can be used in very special cases; so they can not be used to solve equations of numerous realistic scenarios. He's variational iteration method is a powerful device for solving functional equations. This scheme has been used for fractional partial differential equations [3], for solving Helmholtz equation [4], for Burger and coupled Burger's equations [5]. Also authors of this paper have employed variational iteration method for solving fourth order parabolic equations [6], linear and nonlinear systems of ordinary differential equations [7], wave equations with initial and boundary conditions [8], and hyperbolic differential equations [9]. Wazwaz has used this method for rational solutions for KdV, K(2, 2), Burger, and cubic Boussinesq equations [10], and for solving singular initial value problems in the second order differential equations [11].

The variational iteration method [12–16] is in fact a modification of general Lagrange multiplier method [17]. To illustrate the method, consider the following nonlinear equation

$$Lu(t) + Nu(t) = g(t) \quad (1)$$

where L is a linear operator, N is a nonlinear operator, and $g(t)$ is a known analytical function. According to variational iteration method, we can construct the following correction functional.

$$u_{n+1}(t) = u_n(t) + \int_0^t \lambda(\xi) (Lu_n(\xi) + N\tilde{u}_n(\xi) - g(\xi)) d\xi \quad (2)$$

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where λ is general Lagrange multiplier which can be identified via variational theory. $u_0(t)$ is an initial approximation with possible unknowns, and $\tilde{u}_n$ is considered as restricted variation [18], i.e. $\delta\tilde{u}_n = 0$. Therefore, we first determine the Lagrange multiplier λ that will be identified optimally via integration by parts. The successive approximations $u_{n+1}(t)$ for the solution $u(t)$ will be readily obtained upon using the obtained Lagrange multiplier and by using any selective function u_0 . Consequently, the exact solution may be obtained by using $u = \lim_{n \rightarrow \infty} u_n$.

2. Analysis of the method

2.1. One-dimensional Burger's equation

Consider the following one-dimensional Burger's equation with the following initial and boundary conditions;

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2}. \quad (3)$$

Initial condition:

$$u(x, 0) = f(x), \quad 0 \leq x \leq l. \quad (4)$$

Boundary conditions:

$$u(0, t) = f_1(t), \quad \frac{\partial u}{\partial x}(0, t) = f_2(t), \quad t > 0 \quad (5)$$

where $u = u(x, t)$ is the unknown function we are looking for in some domain, v is a parameter ($v > 0$), and $u \frac{\partial u}{\partial x}$ is the nonlinear term.

For solving Eq. (3) with initial condition (4), via VIM, its correction functional can be written as follows:

$$u_{n+1}(x, t) = u_n(x, t) + \int_0^t \lambda(\xi) \left(\frac{\partial u_n}{\partial \xi}(x, \xi) + \tilde{u}_n \frac{\partial \tilde{u}_n}{\partial x}(x, \xi) - v \frac{\partial^2 \tilde{u}_n}{\partial x^2}(x, \xi) \right) d\xi.$$

To make this correction functional stationary, having $\delta u_n(x, 0) = 0$, we derive:

$$\delta u_{n+1}(x, t) = \delta u_n(x, t) + \int_0^t \lambda(\xi) (\delta u_n(x, \xi))' d\xi = 0.$$

Its stationary conditions can be determined as follows:

$$\begin{aligned} \lambda'(\xi) &= 0, \\ 1 + \lambda(\xi)|_{\xi=t} &= 0. \end{aligned}$$

From which the Lagrange multiplier can be identified as $\lambda = -1$, and the following iteration formula is obtained,

$$u_{n+1}(x, t) = u_n(x, t) - \int_0^t \left(\frac{\partial u_n}{\partial \xi}(x, \xi) + u_n \frac{\partial u_n}{\partial x}(x, \xi) - v \frac{\partial^2 u_n}{\partial x^2}(x, \xi) \right) d\xi. \quad (6)$$

Beginning with $u_0 = u(x, 0) = f(x)$, the approximate solution of (3) can be determined by iterative formula (6).

Similarly, to solve Eq. (3) with boundary conditions (5) by VIM we have:

$$u_{n+1}(x, t) = u_n(x, t) + \int_0^x \lambda(\eta) \left(\frac{\partial \tilde{u}_n}{\partial t}(\eta, t) + \tilde{u}_n \frac{\partial \tilde{u}_n}{\partial \eta}(\eta, t) - v \frac{\partial^2 u_n}{\partial \eta^2}(\eta, t) \right) d\eta.$$

To find optimal value of λ , having $\delta u_n(0, t) = 0$, leads to

$$\delta u_{n+1}(x, t) = \delta u_n(x, t) + \delta \int_0^x \lambda(\eta) \left(\frac{\partial \tilde{u}_n}{\partial t}(\eta, t) + \tilde{u}_n \frac{\partial \tilde{u}_n}{\partial \eta}(\eta, t) - v \frac{\partial^2 u_n}{\partial \eta^2}(\eta, t) \right) d\eta,$$

This yields:

$$\delta u_{n+1}(x, t) = \delta u_n(x, t) - v \lambda(\eta) \delta u_n'(\eta, t) \Big|_0^x - \lambda'(\eta) (\delta u_n(\eta, t)) \Big|_0^x + \int_0^x \lambda''(\eta) \delta u_n(\eta, t) d\eta = 0.$$

Therefore, the stationary conditions are obtained as:

$$\begin{aligned} \lambda''(\eta) &= 0, \\ 1 + v \lambda'(\eta) \Big|_{\eta=x} &= 0, \\ \lambda(\eta) \Big|_{\eta=x} &= 0. \end{aligned}$$

This results in:

$$\lambda(\eta) = \frac{1}{v}(x - \eta).$$

And desired iterative relation can be constructed as:

$$u_{n+1}(x, t) = u_n(x, t) + \frac{1}{v} \int_0^x (x - \eta) \left(\frac{\partial u_n}{\partial t}(\eta, t) + u_n \frac{\partial u_n}{\partial \eta}(\eta, t) - v \frac{\partial^2 u_n}{\partial \eta^2}(\eta, t) \right) d\eta. \quad (7)$$

Beginning with $u_0 = f_1(t) + xf_2(t)$, the approximate solution of (3) can be determined via iterative formula (7).

2.2. Two-dimensional Burger's equations

Consider the following system of two-dimensional Burger's equations [19];

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= \frac{1}{R} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= \frac{1}{R} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right). \end{aligned} \quad (8)$$

Subject to initial conditions,

$$\begin{aligned} u(x, y, 0) &= f(x, y), \quad (x, y) \in D, \\ v(x, y, 0) &= g(x, y), \quad (x, y) \in D. \end{aligned} \quad (9)$$

And boundary conditions,

$$\begin{aligned} u(x, y, t) &= f_1(x, y, t), \quad x, y \in \partial D, \quad t > 0, \\ v(x, y, t) &= f_2(x, y, t), \quad x, y \in \partial D, \quad t > 0 \end{aligned} \quad (10)$$

where $D = \{(x, y) | a \leq x \leq b, a \leq y \leq b\}$ and ∂D is its boundary, $u(x, y, t)$ and $v(x, y, t)$ are the velocity components to be determined, f, g, f_1 , and f_2 are known functions, and R is the Reynolds number.

To solve system (8) with initial condition (9), we can construct the following correction functionals:

$$\begin{aligned} u_{n+1}(x, y, t) &= u_n(x, y, t) \\ &+ \int_0^t \lambda_1(\tau) \left[\frac{\partial u_n}{\partial \tau}(x, y, \tau) + \tilde{u}_n \frac{\partial \tilde{u}_n}{\partial x}(x, y, \tau) + \tilde{v}_n \frac{\partial \tilde{u}_n}{\partial y}(x, y, \tau) - \frac{1}{R} \left(\frac{\partial^2 \tilde{u}_n}{\partial x^2}(x, y, \tau) + \frac{\partial^2 \tilde{u}_n}{\partial y^2}(x, y, \tau) \right) \right] d\tau, \\ v_{n+1}(x, y, t) &= v_n(x, y, t) \\ &+ \int_0^t \lambda_2(\tau) \left[\frac{\partial v_n}{\partial \tau}(x, y, \tau) + \tilde{u}_n \frac{\partial \tilde{v}_n}{\partial x}(x, y, \tau) + \tilde{v}_n \frac{\partial \tilde{v}_n}{\partial y}(x, y, \tau) - \frac{1}{R} \left(\frac{\partial^2 \tilde{v}_n}{\partial x^2}(x, y, \tau) + \frac{\partial^2 \tilde{v}_n}{\partial y^2}(x, y, \tau) \right) \right] d\tau. \end{aligned}$$

Following the analysis presented above, the stationary conditions can be obtained as follows:

$$\begin{aligned} \lambda'_1(\tau) &= \lambda'_2(\tau) = 0, \\ 1 + \lambda_1(\tau)|_{\tau=t} &= 1 + \lambda_2(\tau)|_{\tau=t} = 0. \end{aligned}$$

The Lagrange multipliers can be identified as follows;

$$\lambda_1(\tau) = \lambda_2(\tau) = -1.$$

Let's take initial conditions, as initial approximations:

$$u_0(x, y, t) = f(x, y), \quad v_0(x, y, t) = g(x, y).$$

The desired iterative scheme is found as follows:

$$\begin{aligned} u_{n+1}(x, y, t) &= u_n(x, y, t) \\ &- \int_0^t \left[\frac{\partial u_n}{\partial \tau}(x, y, \tau) + u_n \frac{\partial u_n}{\partial x}(x, y, \tau) + v_n \frac{\partial u_n}{\partial y}(x, y, \tau) - \frac{1}{R} \left(\frac{\partial^2 u_n}{\partial x^2}(x, y, \tau) + \frac{\partial^2 u_n}{\partial y^2}(x, y, \tau) \right) \right] d\tau, \\ v_{n+1}(x, y, t) &= v_n(x, y, t) \\ &- \int_0^t \left[\frac{\partial v_n}{\partial \tau}(x, y, \tau) + u_n \frac{\partial v_n}{\partial x}(x, y, \tau) + v_n \frac{\partial v_n}{\partial y}(x, y, \tau) - \frac{1}{R} \left(\frac{\partial^2 v_n}{\partial x^2}(x, y, \tau) + \frac{\partial^2 v_n}{\partial y^2}(x, y, \tau) \right) \right] d\tau. \end{aligned} \quad (11)$$

3. Examples

In this part four examples are provided to illustrate the method, its effectiveness and simplicity.

Example 1. Consider the following initial value equation [20]:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2},$$

$$u(x, 0) = 2x, \quad t > 0.$$

For this example, iterative formula (6), can be written as

$$u_{n+1}(x, t) = u_n(x, t) - \int_0^t \left(\frac{\partial u_n}{\partial \xi} + u_n \frac{\partial u_n}{\partial x} - \frac{\partial^2 u_n}{\partial x^2} \right) d\xi. \quad (12)$$

Starting with $u_0(x, t) = 2x$, the following results can be derived from iterative formula (12),

$$\begin{aligned} u_1(x, t) &= 2x - 4xt, \\ u_2(x, t) &= 2x - 4xt + 8xt^2 - \frac{16}{3}xt^3, \\ u_3(x, t) &= 2x - 4xt + 8xt^2 - 16xt^3 + \frac{64}{3}xt^4 - \frac{64}{3}xt^5 + \frac{128}{9}xt^6 - \frac{256}{63}xt^7, \\ u_4(x, t) &= 2x - 4xt + 8xt^2 - 16xt^3 + 32xt^4 - \frac{832}{15}xt^5 + \frac{256}{3}xt^6 - \frac{7424}{63}xt^7 + \frac{9088}{63}xt^8 \\ &\quad - \frac{88064}{567}xt^9 + \frac{45056}{315}xt^{10} - \frac{20480}{189}xt^{11} + \frac{4096}{63}xt^{12} - \frac{16384}{567}xt^{13} + \frac{32768}{3969}xt^{14} - \frac{65536}{59536}xt^{15}, \\ u_5(x, t) &= 2x - 4xt + 8xt^2 - 16xt^3 + 32xt^4 - 64xt^5 + \frac{5504}{45}xt^6 - \frac{3328}{15}xt^7 + \frac{120704}{315}xt^8 + \dots, \\ &\vdots \end{aligned}$$

Thus, we have:

$$\begin{aligned} u(x, t) &= \lim_{n \rightarrow \infty} u_n(x, t) \\ &= 2x(1 - 2t + 4t^2 - 8t^3 + 16t^4 - \dots) \\ &= \sum_{n=0}^{\infty} (-1)^n 2^{n+1} xt^n \\ &= \frac{2x}{1 + 2t}. \end{aligned}$$

This is an exact solution.

Example 2. Consider the following partial differential equation [21]:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2},$$

$$u(0, t) = 0, \quad \frac{\partial u}{\partial x}(0, t) = \frac{1}{t} - \frac{\pi^2}{2vt^2}, \quad t > 0.$$

He's variational iteration method leads to the following iterative scheme:

$$u_{n+1}(x, t) = u_n(x, t) + \frac{1}{v} \int_0^x (x - \eta) \left(\frac{\partial u_n(\eta, t)}{\partial t} + u_n(\eta, t) \frac{\partial u_n(\eta, t)}{\partial \eta} - v \frac{\partial^2 u_n(\eta, t)}{\partial \eta^2} \right) d\eta \quad (13)$$

Starting with $u_0(x, t) = \left(\frac{1}{t} - \frac{\pi^2}{2vt^2} \right) x$, and using iterative formula (13), the following results are obtained:

$$\begin{aligned} u_1(x, t) &= \frac{x}{t} - \frac{\pi^2 x}{2vt^2} + \frac{\pi^4 x^3}{24v^3 t^4}, \\ u_2(x, t) &= \frac{x}{t} - \frac{\pi^2 x}{2vt^2} + \frac{\pi^4 x^3}{24v^3 t^4} - \frac{\pi^6 x^5}{240v^5 t^6} + \frac{\pi^6 x^7}{8064v^7 t^8}, \end{aligned}$$

$$\begin{aligned}
u_3(x, t) &= \frac{x}{t} - \frac{\pi^2 x}{2\nu t^2} + \frac{\pi^4 x^3}{24\nu^3 t^4} - \frac{\pi^6 x^5}{240\nu^5 t^6} + \frac{17\pi^8 x^7}{40320\nu^7 t^8} - \frac{19\pi^{10} x^9}{725760\nu^9 t^{10}} \\
&\quad + \frac{67\pi^{12} x^{11}}{53222400\nu^{11} t^{12}} - \frac{19\pi^{14} x^{13}}{25159680\nu^{13} t^{14}} + \frac{19\pi^{16} x^{15}}{1950842880\nu^{15} t^{16}}, \\
u_4(x, t) &= \frac{x}{t} - \frac{\pi^2 x}{2\nu t^2} + \frac{\pi^4 x^3}{24\nu^3 t^4} - \frac{\pi^6 x^5}{240\nu^5 t^6} + \frac{17\pi^8 x^7}{40320\nu^7 t^8} - \frac{31\pi^{10} x^9}{725760\nu^9 t^{10}} \\
&\quad + \frac{571\pi^{12} x^{11}}{159667200\nu^{11} t^{12}} - \frac{3331\pi^{14} x^{13}}{1245041600\nu^{13} t^{14}} + \frac{377017\pi^{16} x^{15}}{2092278988800\nu^{15} t^{16}} + \dots, \\
&\vdots
\end{aligned}$$

Therefore, we have:

$$\begin{aligned}
u(x, t) &= \lim_{n \rightarrow \infty} u_n(x, t) \\
&= \frac{x}{t} - \frac{\pi^2 x}{2\nu t^2} + \frac{\pi^4 x^3}{24\nu^3 t^4} - \frac{\pi^6 x^5}{240\nu^5 t^6} + \frac{17\pi^8 x^7}{40320\nu^7 t^8} - \frac{31\pi^{10} x^9}{725760\nu^9 t^{10}} + \dots \\
&= \frac{x}{t} - \frac{\pi}{t} \left(\frac{\pi x}{2\nu t} - \frac{\pi^3 x^3}{3(2)^3 \nu^3 t^3} + \frac{2\pi^5 x^5}{15(2)^5 \nu^5 t^5} - \frac{17\pi^7 x^7}{315(2)^7 \nu^7 t^7} + \frac{62\pi^9 x^9}{2835(2)^9 \nu^9 t^9} - \dots \right) \\
&= \frac{x}{t} - \frac{\pi}{t} \tanh\left(\frac{\pi x}{2\nu t}\right),
\end{aligned}$$

This is also an exact solution.

It's worth pointing out that the results of Examples 1 and 2 are exactly the same as the result of applying Adomian decomposition method [21,20]. As an advantage of Variational iteration method over Adomian decomposition procedure, the former method provides the solution without calculating Adomian's polynomials. This technique solves the problem without any need for discretization of the variables.

Example 3. Consider the two-dimensional Burger's equation (8), with the following initial conditions:

$$u(x, y, 0) = x + y,$$

$$v(x, y, 0) = x - y.$$

Via He's variational iteration method, we have:

$$\begin{aligned}
u_{n+1}(x, y, t) &= u_n(x, y, t) - \int_0^t \left[\frac{\partial u_n}{\partial \tau} + u_n \frac{\partial u_n}{\partial x} + v_n \frac{\partial u_n}{\partial y} - \frac{1}{R} \frac{\partial^2 u_n}{\partial x^2} - \frac{1}{R} \frac{\partial^2 u_n}{\partial y^2} \right] d\tau, \\
v_{n+1}(x, y, t) &= v_n(x, y, t) - \int_0^t \left[\frac{\partial v_n}{\partial \tau} + u_n \frac{\partial v_n}{\partial x} + v_n \frac{\partial v_n}{\partial y} - \frac{1}{R} \frac{\partial^2 v_n}{\partial x^2} - \frac{1}{R} \frac{\partial^2 v_n}{\partial y^2} \right] d\tau.
\end{aligned} \tag{14}$$

Considering the initial approximations $u_0(x, y, t) = x + y$, $v_0(x, y, t) = x - y$, and applying Variational iteration formula (14), other terms of the sequence are computed as follows:

$$u_1(x, t) = x + y - 2xt$$

$$v_1(x, t) = x - y - 2yt$$

$$u_2(x, t) = x + y - 2xt + 2xt^2 + 2yt^2 - \frac{4}{3}xt^3,$$

$$v_2(x, t) = x - y - 2yt + 2xt^2 - 2yt^2 - \frac{4}{3}yt^3,$$

$$u_3(x, t) = x + y - 2xt + 2xt^2 + 2yt^2 - 4xt^3 + \frac{8}{3}xt^4 + \frac{8}{3}yt^4 - \frac{8}{3}xt^5 + \frac{8}{9}xt^6 + \frac{8}{9}yt^6 - \frac{16}{63}xt^7,$$

$$v_3(x, t) = x - y - 2yt + 2xt^2 - 2yt^2 - 4yt^3 + \frac{8}{3}xt^4 + \frac{8}{3}yt^4 - \frac{8}{3}yt^5 + \frac{8}{9}xt^6 - \frac{8}{9}yt^6 - \frac{16}{63}yt^7,$$

$$u_4(x, t) = x + y - 2xt + 2xt^2 + 2yt^2 - 4xt^3 + 4xt^4 + 4yt^4 - \frac{104}{5}xt^5 + \frac{48}{9}xt^6 + \frac{48}{9}yt^6 + \dots,$$

$$v_4(x, t) = x - y - 2yt + 2xt^2 - 2yt^2 - 4yt^3 + 4xt^4 - 4yt^4 - \frac{104}{5}yt^5 + \frac{48}{9}xt^6 - \frac{48}{9}yt^6 - \dots,$$

$\vdots$

Table 1The VIM results for $u(x, y, t)$ for the first five approximations with $R = 1$.

(x, y)	$t = 0.01$		$t = 0.5$		$t = 2$	
	$u^*(x, y, t)$	$ u^* - u $	$u^*(x, y, t)$	$ u^* - u $	$u^*(x, y, t)$	$ u^* - u $
(0.1, 0.1)	0.6249804684	3.0E–10	0.6240234569	5.0E–10	0.6210950204	7.0E–10
(0.5, 0.1)	0.6218561399	8.0E–9	0.6208999164	7.5E–9	0.6179761639	7.5E–9
(0.9, 0.1)	0.6187357218	1.0E–9	0.6177814784	6.0E–10	0.6148660410	8.0E–10
(0.3, 0.3)	0.6249804684	3.0E–10	0.6240234569	5.0E–10	0.6210950204	7.0E–10
(0.7, 0.3)	0.6218561399	8.0E–9	0.6208999164	7.5E–9	0.6179761639	7.5E–9
(0.1, 0.5)	0.6281048352	5.2E–9	0.6271482314	5.4E–9	0.6242187657	5.5E–9
(0.5, 0.5)	0.6249804684	3.0E–10	0.6240234569	5.0E–10	0.6210950204	7.0E–10
(0.9, 0.5)	0.6218561399	8.0E–9	0.6240234569	5.0E–10	0.6179761639	7.5E–9

Table 2The VIM results for $v(x, y, t)$ for the first five approximations with $R = 1$.

(x, y)	$t = 0.01$		$t = 0.5$		$t = 2$	
	$v^*(x, y, t)$	$ v^* - v $	$v^*(x, y, t)$	$ v^* - v $	$v^*(x, y, t)$	$ v^* - v $
(0.1, 0.1)	0.8750195310	3.0E–10	0.8759765418	8.0E–10	0.8789049782	7.0E–10
(0.5, 0.1)	0.8781438776	9.5E–9	0.8791001011	1.0E–8	0.8820238539	1.0E–8
(0.9, 0.1)	0.8812642805	1.3E–9	0.8822185238	1.6E–9	0.8851339616	1.8E–9
(0.3, 0.3)	0.8750195310	3.0E–10	0.8759765418	8.0E–10	0.8789049782	7.0E–10
(0.7, 0.3)	0.8781438776	9.5E–9	0.8791001011	1.0E–8	0.8820238539	1.0E–8
(0.1, 0.5)	0.8718951772	7.2E–9	0.8728517809	6.9E–9	0.8757812466	6.8E–9
(0.5, 0.5)	0.8750195310	3.0E–10	0.8759765418	8.0E–10	0.8789049782	7.0E–10
(0.9, 0.5)	0.8781438776	9.5E–9	0.8759765418	8.0E–10	0.8820238539	1.0E–8

Therefore,

$$\begin{aligned}
 u(x, y, t) &= \lim_{n \rightarrow \infty} u_n(x, y, t) \\
 &= x + y - 2xt + 2xt^2 + 2yt^2 - 4xt^3 + 4xt^4 + 4yt^4 - 8xt^5 + \dots \\
 &= x(1 + 2t^2 + 4t^4 + \dots) + y(1 + 2t^2 + 4t^4 + \dots) - 2xt(1 + 2t^2 + 4t^4 + \dots) \\
 &= \frac{x + y - 2xt}{1 - 2t^2}
 \end{aligned}$$

and

$$\begin{aligned}
 v(x, y, t) &= \lim_{n \rightarrow \infty} v_n(x, y, t) \\
 &= x - y - 2yt + 2xt^2 - 2yt^2 - 4yt^3 + 4xt^4 - 4yt^4 - 8yt^5 + \dots \\
 &= x(1 + 2t^2 + 4t^4 + \dots) - y(1 + 2t^2 + 4t^4 + \dots) - 2yt(1 + 2t^2 + 4t^4 + \dots) \\
 &= \frac{x - y - 2yt}{1 - 2t^2}
 \end{aligned}$$

which are exact solutions.

Example 4. Let's consider system of Burger's equations (8), with the following initial conditions [19]:

$$\begin{aligned}
 u(x, y, 0) &= \frac{3}{4} - \frac{1}{4[1 + \exp(y - x)R/8]}, \\
 v(x, y, 0) &= \frac{3}{4} + \frac{1}{4[1 + \exp(y - x)R/8]}.
 \end{aligned} \tag{15}$$

With the exact solutions:

$$\begin{aligned}
 u(x, y, t) &= \frac{3}{4} - \frac{1}{4[1 + \exp(4y - 4x - t)R/32]}, \\
 v(x, y, t) &= \frac{3}{4} + \frac{1}{4[1 + \exp(4y - 4x - t)R/32]}.
 \end{aligned}$$

To solve system (8) by VIM, we follow the same procedure discussed in example (3) and obtain the iterative relations (14); in this example initial approximations (15) are considered. The accuracy of VIM for the system of two-dimensional Burger's equation agrees very well with the exact solutions, and absolute errors are very small for the present choice of x, y , and t . These results are shown in Tables 1–4 for $R = 1$ and $R = 10$.

Table 3The VIM results for $v(x, y, t)$ for the first five approximations with $R = 10$.

(x, y)	$t = 0.01$		$t = 0.5$		$t = 2$	
	$v^*(x, y, t)$	$ v^* - v $	$v^*(x, y, t)$	$ v^* - v $	$v^*(x, y, t)$	$ v^* - v $
(0.1, 0.1)	0.6248046876	1.0E–10	0.6152542254	3.1E–8	0.5873095999	1.4E–5
(0.5, 0.1)	0.5942016362	4.7E–9	0.5853956315	6.6E–9	0.5613388416	6.7E–5
(0.9, 0.1)	0.5670818670	3.8E–9	0.5598373321	4.7E–9	0.5410918610	3.7E–5
(0.3, 0.3)	0.6248046876	1.0E–10	0.6152542254	3.1E–8	0.5873095999	1.4E–5
(0.7, 0.3)	0.5942016362	4.7E–9	0.5853956315	6.6E–9	0.5613388416	6.7E–5
(0.1, 0.5)	0.6554311623	3.8E–9	0.6462752733	1.5E–8	0.6171168308	8.1E–5
(0.5, 0.5)	0.6248046876	1.0E–10	0.6152542254	3.1E–8	0.5873095999	1.4E–5
(0.9, 0.5)	0.5942016362	4.7E–9	0.5853956315	6.6E–9	0.5613388416	6.7E–5

Table 4The VIM results for $v(x, y, t)$ for the first five approximations with $R = 10$.

(x, y)	$t = 0.01$		$t = 0.5$		$t = 2$	
	$v^*(x, y, t)$	$ v^* - v $	$v^*(x, y, t)$	$ v^* - v $	$v^*(x, y, t)$	$ v^* - v $
(0.1, 0.1)	0.8751953122	1.0E–10	0.8847457745	3.1E–8	0.9126903999	1.5E–4
(0.5, 0.1)	0.9057983514	7.7E–9	0.9146043563	1.9E–8	0.9386611460	6.8E–5
(0.9, 0.1)	0.9329181417	4.9E–9	0.9401626777	1.4E–8	0.9589081486	3.7E–5
(0.3, 0.3)	0.8751953122	1.0E–10	0.8847457745	3.1E–8	0.9126903999	1.5E–4
(0.7, 0.3)	0.9057983514	7.7E–9	0.9146043563	1.9E–8	0.9386611460	6.8E–5
(0.1, 0.5)	0.8445688292	4.7E–9	0.8537247185	7.1E–9	0.8828831610	8.1E–5
(0.5, 0.5)	0.8751953122	1.0E–10	0.8847457745	3.1E–8	0.9126903999	1.5E–4
(0.9, 0.5)	0.9057983514	7.7E–9	0.9146043563	1.9E–8	0.9386611460	6.8E–5

4. Conclusions

In this paper, variational iteration method is proposed for solving Burger's equation. He's variational iteration method is suitable as an alternative approach to current techniques being employed to a wide variety of problems in physics. The small amount of computation compared to that required in numerical methods [19], and the rapid convergence show that the method is reliable and provides a significant improvement in solving partial differential equations over existing methods. The main advantage of the VIM over A.D.M. is that this method provides the solution without a need for calculating Adomian's polynomials [21,20]. Computations in this paper are performed using Maple 10.

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